Google Developer Group
Dubai

Introduction To ADK 2.0

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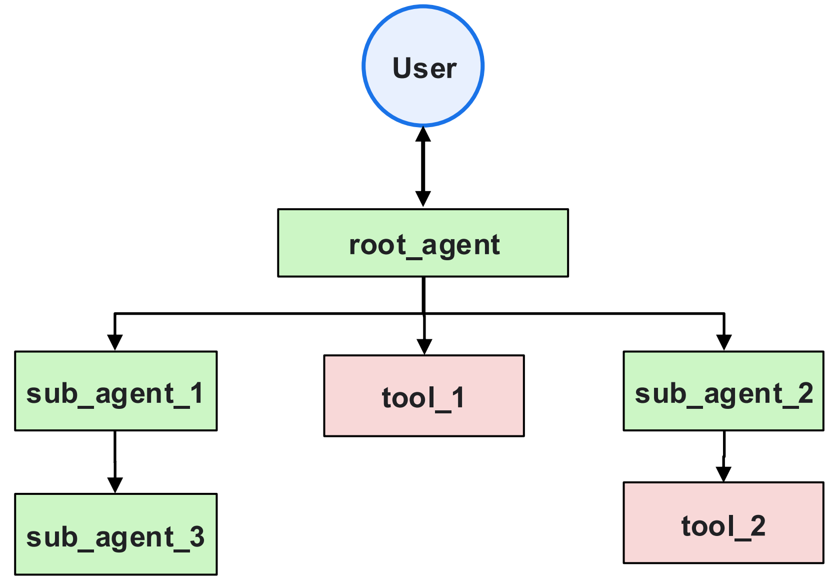
{ Build with AI }

ADK 1.x - Mental Model

- One **root_agent** talks to the user
- **sub_agents** = specialists (intake, legal, security, manager)
- Coordinator **transfers** via prompts when the LLM decides
- Pipeline order is **implicit**, not declared
- Heavily relies on **prompting** and instruction following



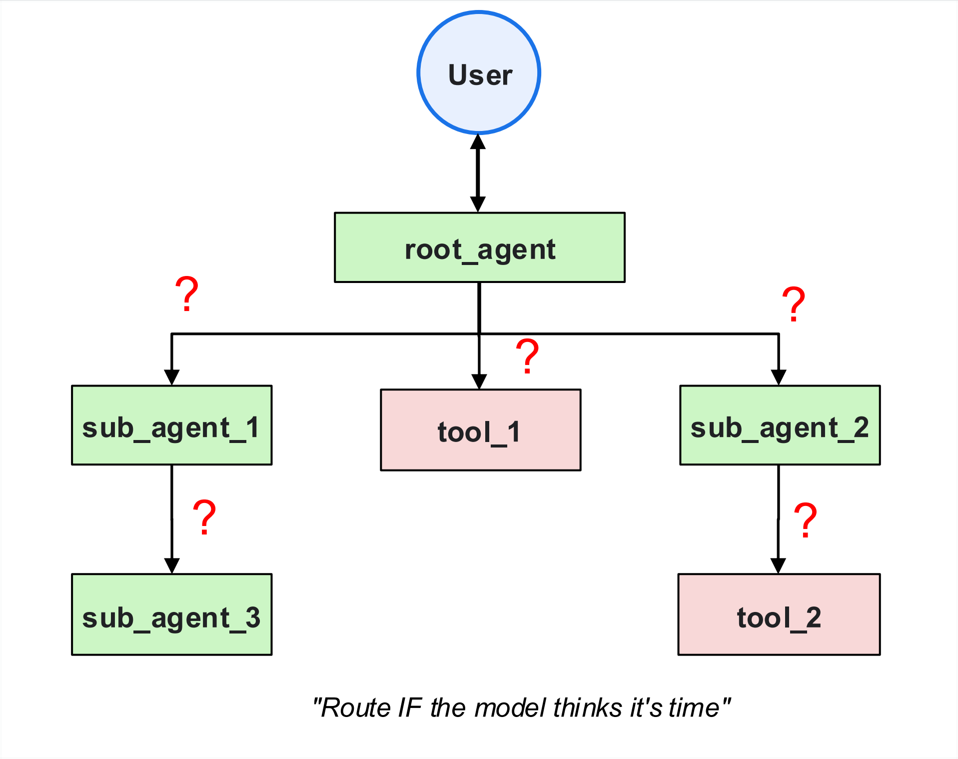
"Route when the model thinks it's time"

ADK 1.x - What Worked

- Fast conversational multi-agent prototypes
- Non-deterministic flow
  - User may ask anything on topic and agent can “figure it out”
- sub_agents + tools still an option in 2.0 (collaborative path)
- Strong fit: delegation-first UX
  - “ask my coordinator”

ADK 1.x - Push To Upgrade

- No pipeline diagram:
  - Less deterministic
  - Hard to review in UI or code review
- Procedure in prompts:
  - Steps can be skipped or reordered
- Parallelism unclear:
  - “Run both tools in parallel” is hope, not structure
- Human-In-The-Loop:
  - Possible, but wiring is ad-hoc



ADK 2.0 - Introducing 3 New Paradigms

Graph Workflows

- Deterministic Pipelines
- Declare flow in edges and use Event routing for branches
- Concrete, limited execution flows

Dynamic Workflows

- Algorithmic Control
- Implement control flow in Python inside `@node` orchestrators
- Run-time based branching

Collaborative Workflows

- Agentic Delegation
- Coordinator delegates to specialized sub-agents via tools
- Multi-agent product shapes for open-ended questions

ADK 2.0 - Introducing 3 New Paradigms

Paradigm	When To Use It	Ideal Enterprise Task
Graph Workflows	Strict SOPs, compliance-heavy routing, predictable execution.	**Automated Loan/Credit Processing:** Parse application ➔ Check credit API ➔ Route to Approve/Deny.
Dynamic Workflows	Algorithmic logic, unknown loop iterations, runtime-shaped branching.	**Automated Code Refactoring:** Loop through files ➔ Generate tests ➔ Re-prompt until tests pass.
Collaborative Workflows	Open-ended problem solving, multi-persona delegation, conversational routing.	**Autonomous Content Team:** Lead Agent assigns research, drafting, and compliance checks to sub-agents.

Case Study: AI Procurement Agent

Problem Statement & Requirements

- Automate manual software license and subscription requests
- Gather annual cost in AED and business justification from users
- Execute mandatory security and legal compliance checks
- Auto-approve low-value subscriptions (<500 AED)
- Trigger manager workflow for high-value procurement

Follow The Code!

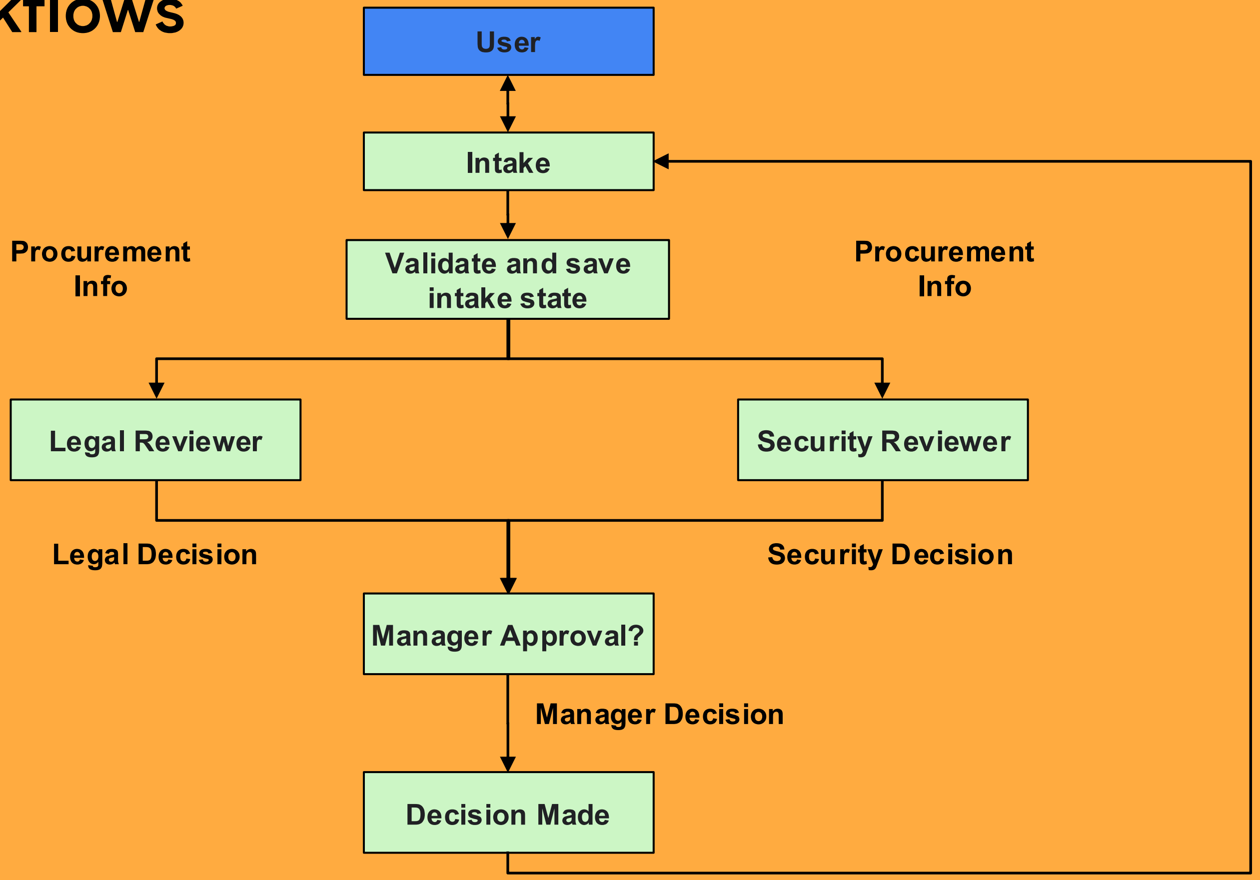
Github code - Clone locally

github.com/ro1406/adk-2.0-tutorial

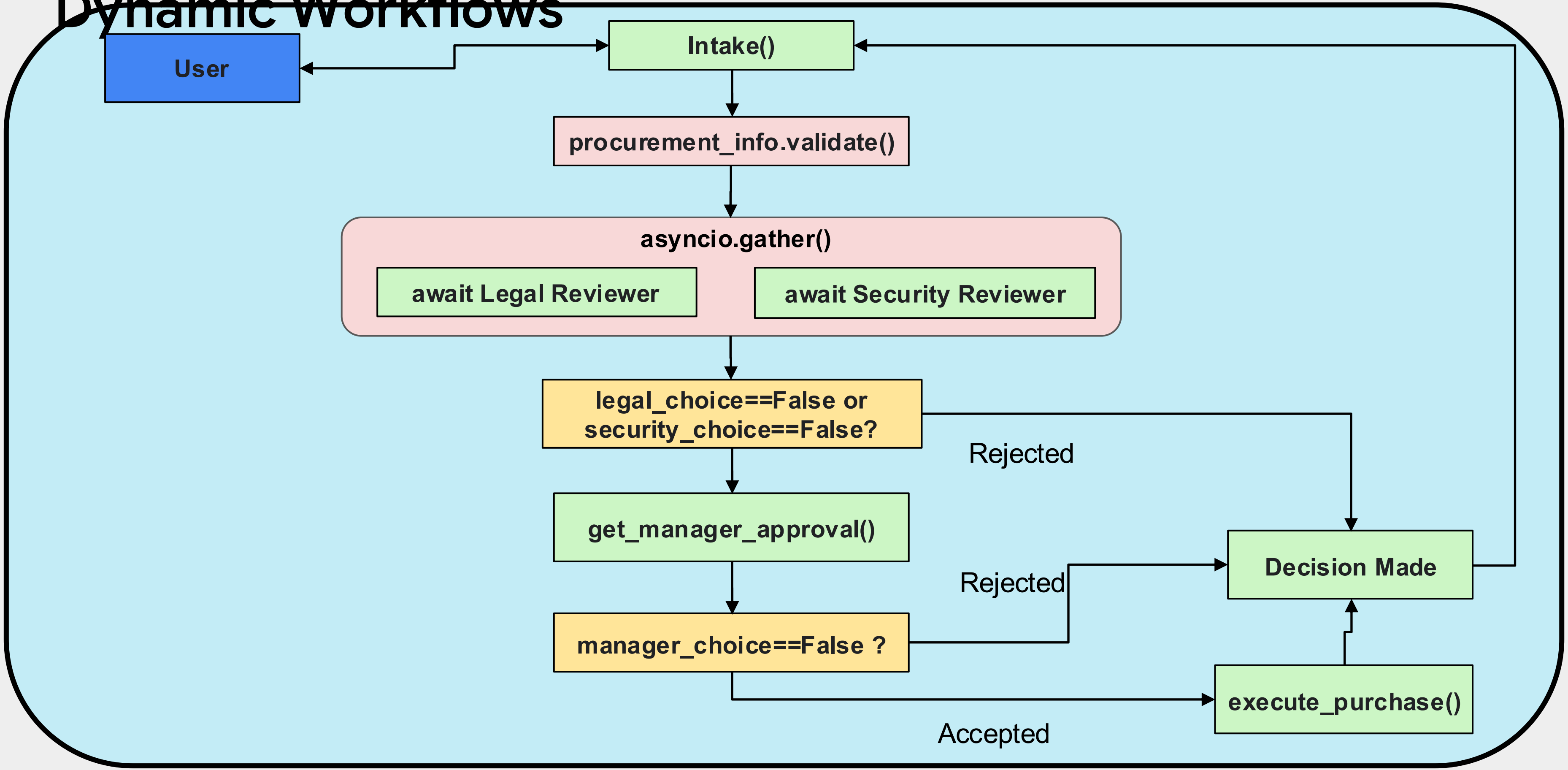
AI Studio - Create Free API Key

https://aistudio.google.com/api-keys

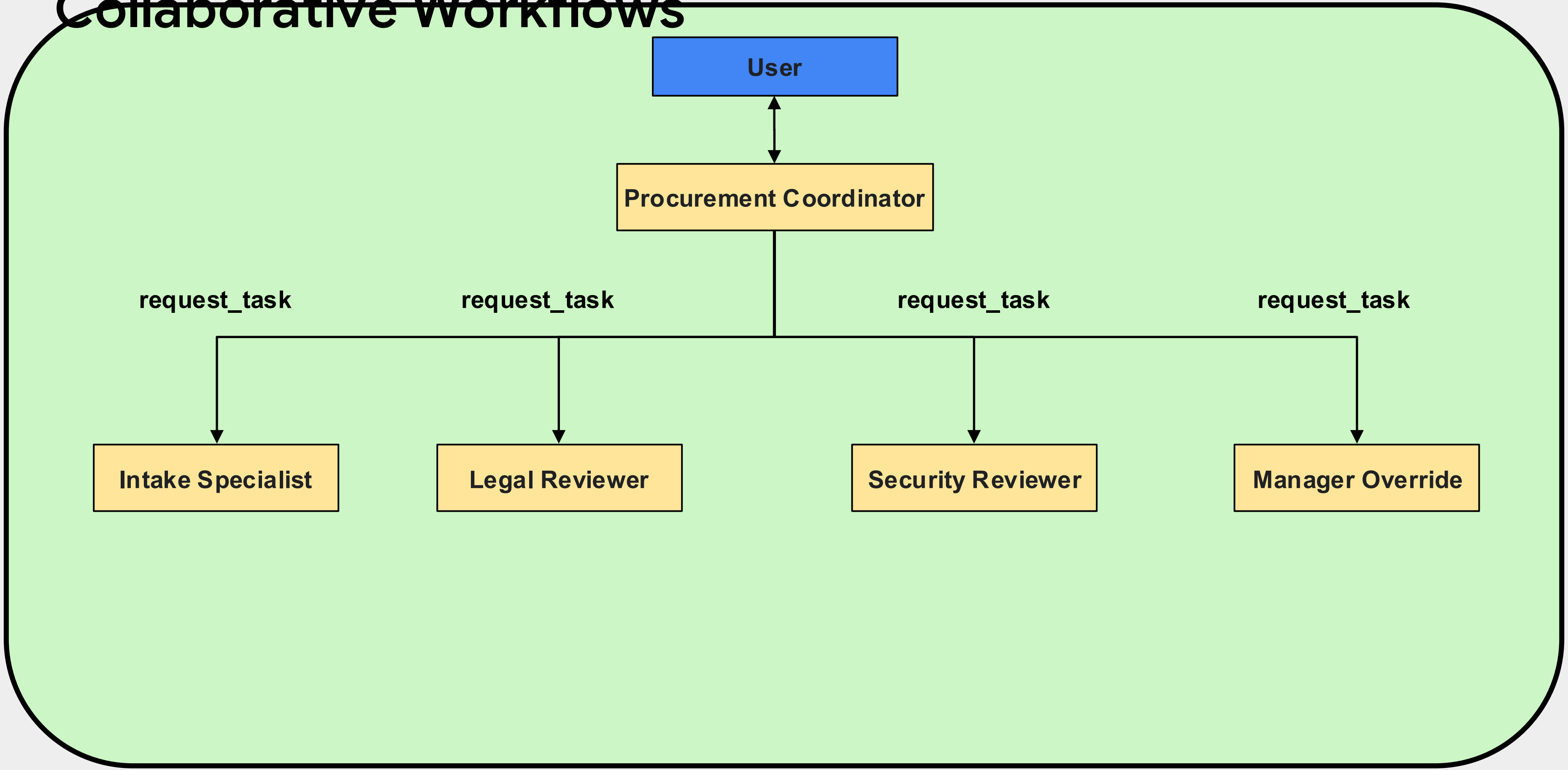
Graph Workflows



Dynamic Workflows



Collaborative Workflows



Why ADK?

- **State Persistence:** Automatic checkpointing and node retries on failure.
- **Built-in Observability:** Visual Web UI for effortless debugging and tracing.
- **Native HITL:** Built-in require_confirmation to pause/resume workflows—no custom webhooks needed.
- **Managed Concurrency:** Out-of-the-box parallel fan-outs, rate limiting, and token management.

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